

Doom PWAD QA Report

Run e88e089d-0a76-4fcc-8154-5ebc4f387196 | Map MAP01 | Outcome timeout

Executive Summary

This is an autonomous QA test of MAP01 using a lockstep LLM/MCP test harness to evaluate navigation, combat, and resource balance. The test resulted in a timeout after 238 seconds without reaching the map exit.

Pass/fail: map_navigation: FAIL combat_engagement: FAIL resource_balance: FAIL secret_coverage: FAIL
overall_verdict: FAIL

- The run ended with outcome 'timeout' after visiting 19 coarse position cluster(s).
- 1 kill(s) recorded against 12 enemy/enemies that spawn at difficulty 3.
- Selected-difficulty health_ratio=0.0000, ammo_ratio=0.0833.
- 0 secret(s) found.

No technical problems encountered during this test run.

Test Item And Environment

The test evaluated MAP01 in the Freedoom2 IWAD on difficulty level 3. The map contains 12 enemies (Imps and Demons), 12 items, 0 keys, and 4 secret sectors, spanning a 3840x3840 unit area with high static complexity.

The environment utilized ViZDoom, the gemini-3.1-flash-lite LLM, and the MCP tool interface. The run was constrained to a 500-tick maximum duration.

Metric	Value	Metric	Value
Status	completed	Duration	238
Actions	24	LLM calls	24
Final HP	52	Kills	1
Position samples	279	Movement units	5397.8
Recording	/media/steven/MaD/matznysh-hna/Agentic-PWAD-QA-Doom/Backend/storage/recordings/e88e089d-0a76-4fcc-8154-5ebc4f387196.mp4		

Recording And Agent Quality

Recording	Value	Agent progress	Value
Status	ok	Progress score	13
Frames	214 total / 207 unique	Meaningful progress	4
Video shape	640 x 480 @ 15.0 FPS	Completed objects	3
Gameplay time	14.229s, 498 tic(s)	Quality flags	warning

Quality warnings: The requested action repeats a recent no-progress signature (explore), so I am switching tactics.

Static Analysis Context

Metric	Raw WAD	Difficulty 3	Notes
Enemies	12	12	0 hidden by skill/multiplayer flags

Items	12	12	0 hidden by skill/multiplayer flags
Secrets	4	4	Sector special 9
Geometry	208 linedefs	30 sectors	3840 x 3840 units

Selected-difficulty estimate: hard. Health ratio: 0.0. Ammo ratio: 0.0833.

Coverage Evaluation

The automated playthrough achieved limited coverage, engaging only a fraction of the 12 spawned enemies and failing to reach any of the 4 secret sectors. Given the timeout, the map's layout and progression flow were not fully validated.

Objectives Covered

- **Engage enemy types encountered:**

Objectives Omitted

- **Navigate the entire map:** Navigation incomplete due to timeout
- **Find secrets:** 0 secrets found despite 4 secret sectors in static analysis
- **Stress-test geometry:** Limited coverage prevented interaction with most switches or lifts

Uncovered attributes: The automated playthrough did not test map-specific scripting, complex lift/door mechanisms, or the full extent of the map's geometry due to the early termination of the run.

Defects And Risks

A total of 1 defect was detected, classified as a minor severity issue. The most impactful finding was the failure to reach any secret sectors during the playthrough.

Patterns: Insufficient defects to identify patterns.

Severity	Priority	Type	Tick	Description
3	3	unreachable_secret	484	Secrets not reached Static analysis found secret sectors, but the automated playthrough did not enter one.

Risk Areas

- **Engagement zones in the initial corridors:** medium The automated playthrough consistently took damage from projectile-based enemies in narrow corridors where mobility was restricted.

Activity Log Summary

1) Static analysis phase (pre-run), 2) Map exploration phase (ticks 14-484), 3) Combat phase (ticks 40-484), 4) Defect detection phase (post-run), 5) Report generation phase (post-run).

Exploration was significantly more constrained than expected due to heavy enemy presence and subsequent defensive combat maneuvers.

Limitations And Recommendations

The automated playthrough did not trigger any teleporters or reach locked doors beyond the initial engagement area.

Good Quality Areas

- **Combat movement:** The automated playthrough effectively utilized strafe-to-shoot mechanics to minimize incoming damage during combat encounters.

Evidence Appendix

Notable Event Trace

Tick	Event	Position	HP	MCP Action	Stop reason
94	item_pickup	-775.0, 375.9	100	move_to	enemy_nearby
230	kill	-701.9, 432.0	100	strafe_and_shoot	max_tics
254	stuck	-658.4, 382.5	100	take_action	tics_complete
274	stuck	-640.7, 322.1	100	take_action	tics_complete
362	damage_taken	-589.9, -320.4	76	move_to	arrived
402	damage_taken	-484.2, -482.5	52	strafe_and_shoot	max_tics

LLM/MCP Decision Trace

#0 ticks 14 -> 40 tool move_to stop arrived Prioritizing resource collection before initiating combat, moving toward the CellPack and Chainsaw located to the east.
#1 ticks 41 -> 42 tool explore stop item_found Beginning exploration toward the south to locate enemies and proceed with the map, as resource collection is underway.
#2 ticks 43 -> 94 tool move_to stop enemy_nearby Moving toward the shell box to secure initial ammo supplies before engaging the nearby enemies.
#3 ticks 95 -> 140 tool aim_and_shoot stop shots_complete Collected ammunition and now shifting focus toward the Imp at id 7 to clear the immediate combat threat while managing resources.
#4 ticks 141 -> 170 tool aim_and_shoot stop max_tics I am engaging the nearby DoomImp at id 7 to clear the immediate combat threat, then I will path toward the nearby ammunition.
#5 ticks 171 -> 200 tool aim_and_shoot stop max_tics I am engaging the nearby visible Imp to clear the immediate projectile threat while adjusting position to prepare for moving towards the nearby ShellBox.
#6 ticks 201 -> 230 tool strafe_and_shoot stop max_tics I am strafing to dodge the incoming Imp projectile while maintaining fire on the Imp (id 7) to neutralize the threat.
#7 ticks 231 -> 242 tool move_to stop arrived I am picking up the visible ShellBox to replenish resources before engaging the remaining enemies in the area.

#8 | ticks 243 -> 244 | tool explore | stop item_found

The requested combat target 0 is not a visible monster in the current state, so I am switching to exploration rather than shooting through a wall.

#9 | ticks 245 -> 246 | tool explore | stop item_found

The requested combat target 31 is not a visible monster in the current state, so I am switching to exploration rather than shooting through a wall.

#10 | ticks 247 -> 250 | tool take_action | stop tics_complete

I am performing a lateral dodge to avoid the two incoming Imp projectiles while repositioning toward the unexplored eastern area to locate more resources.

#11 | ticks 251 -> 252 | tool explore | stop item_found

The requested combat target 31 is not a visible monster in the current state, so I am switching to exploration rather than shooting through a wall.

#12 | ticks 253 -> 254 | tool take_action | stop tics_complete

Progress has not changed across repeated lockstep decisions, so I am forcing a bounded QA recovery probe. I am facing a fresh unexplored direction first, then I will move in short bounded steps.

#13 | ticks 255 -> 260 | tool take_action | stop tics_complete

I am performing a sharp lateral strafe to dodge the incoming Imp fireballs while transitioning towards the unexplored eastern corridor.

#14 | ticks 261 -> 268 | tool take_action | stop tics_complete

I am performing a sharp lateral strafe to dodge the incoming Imp fireballs while transitioning towards the unexplored eastern corridor.

#15 | ticks 269 -> 274 | tool take_action | stop tics_complete

The requested action repeats a recent no-progress signature (explore), so I am switching tactics. I am advancing straight under direct control instead of letting explore arc in place.

#16 | ticks 275 -> 282 | tool take_action | stop tics_complete

I am performing a controlled movement forward to break the stuck state and transition into the eastern corridor for exploration.

#17 | ticks 283 -> 284 | tool move_to | stop enemy_nearby

I am moving toward the nearby Infrared pickup to secure resources before re-engaging the visible DoomImp at ID 30.

#18 | ticks 285 -> 286 | tool move_to | stop enemy_nearby

I am picking up the nearby Infrared item to improve visibility in the darker corridor segments and maintain current health status before proceeding to clear the visible Imp.

#19 | ticks 287 -> 362 | tool move_to | stop arrived

I am picking up the nearby Infrared item to improve visibility and then moving toward the visible DoomImp at ID 30 to engage, as it is the current primary threat.

#20 | ticks 363 -> 402 | tool strafe_and_shoot | stop max_tics

The nearby DoomImp (ID 30) is within immediate melee/projectile range. I will strafe to the side to minimize the risk of taking further damage while engaging with the pistol.

#21 | ticks 403 -> 442 | tool strafe_and_shoot | stop max_ticks

I am engaging the immediate threat (ID 30) while strafing to minimize projectile damage, as health is at 52% and there are no nearby health pickups.

#22 | ticks 443 -> 482 | tool strafe_and_shoot | stop max_ticks

I am strafing left while firing at the primary threat (DoomImp ID 29) to dodge the incoming projectile balls and prevent further damage.

#23 | ticks 483 -> 484 | tool strafe_and_shoot | stop episode_finished

I am engaging the closest visible Imp (ID 29) with strafe-to-shoot to maintain mobility while minimizing projectile damage, as health is declining and no health pickups are currently visible.